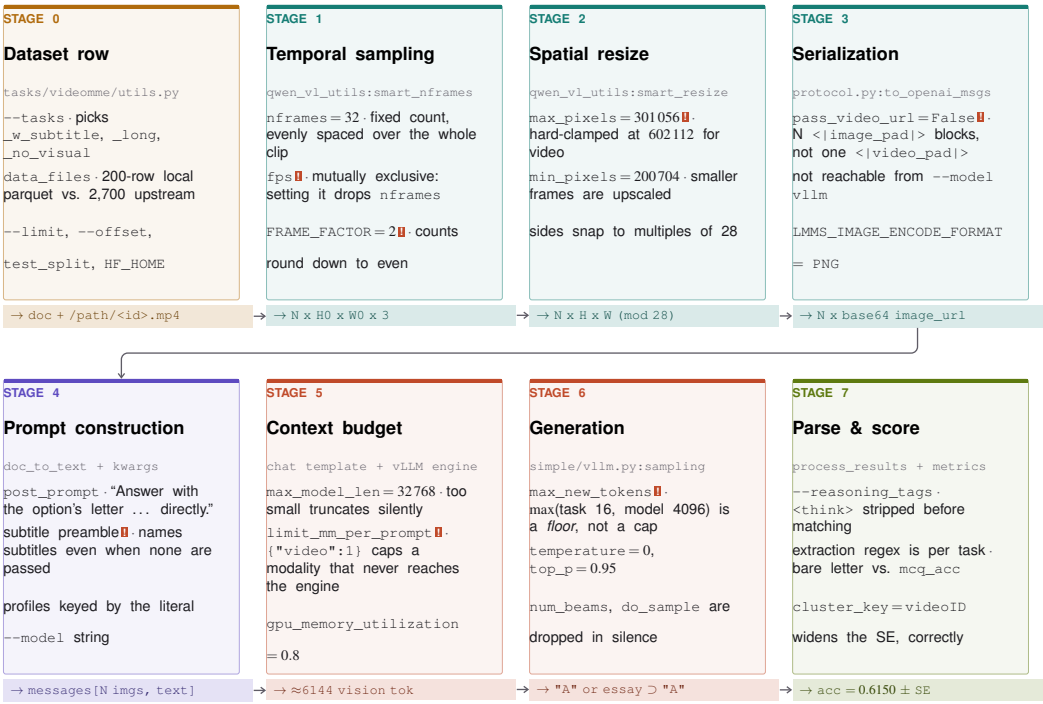


Knobs that do not do what their name suggests. `fps` and `nframes` are mutually exclusive and never merged · `max_frame_num` is read only on the `fps` branch · odd frame counts round down (ask 7, get 6) · `max_pixels` is silently clamped to 602 112 for video frames · `pass_video_url` cannot be reached from `--model vllm` · `max_new_tokens` is a floor, so `--gen_kwargs` alone will not lower it.

Stages 1–2 fix the context bill paid at stage 5. vision_tokens $\approx$ (frames/2) $\times$ (max_pixels/784), where the /2 is the temporal patch merge and $784 = 28 \times 28$ pixels per token after the 2×2 spatial merge; 32 frames at `max_pixels` = 301 056 is $\approx$6,144 vision tokens for a single multiple-choice question.

Figure 1: Everything between a video and a reported accuracy is configuration. A VideoMME row is an `mp4` path, a question, four options and a gold letter; the eight stages above turn it into one number. Each panel lists the settings that act at that stage, the module that reads them, and (in the tinted strip) the representation leaving the stage. Markers ( ) flag settings whose effect does not follow from their name: several are silently ignored, silently clamped, or resolved as a floor rather than a cap. Frame count and per-frame resolution are set entirely from `--model_args`, never from the task YAML, and together they determine the context length paid at stage 5. Values shown are those of our Qwen2.5-VL-7B-Instruct configuration (vLLM 0.11.0, qwen-vl-utils 0.0.14, torchcodec reader).

Reported video-QA accuracies are rarely accompanied by the configuration that produced them. Figure 1 enumerates the settings that a single VideoMME evaluation actually depends on, and Table 1 orders them by how far they move the reported number.

Table 1: Where an unexplained gap between two evaluation runs comes from, ordered by the size of the effect we observe. Stage numbers refer to Figure 1.

#	Knob	Stage	Why it moves the number
1	`nframes` / `fps`	1	Fewer frames carry less temporal evidence. Mutually exclusive: setting `fps` silently disables `nframes`.
2	row set (`--limit`, split, parquet)	0	Different rows are not comparable at all; this checkout evaluates a 200-row subset, not the 2,700-row upstream split.
3	`image_pad` vs. `video_pad`	3	A genuinely different input representation, hence different logits. Not toggleable from `--model vllm`.
4	`max_pixels` / `min_pixels`	2	Per-frame detail; silently clamped to 602 112 for video frames, so the requested value may not be the effective one.
5	`max_new_tokens`	6	Resolved as $\max(\text{task}, \text{model})$, a floor rather than a cap: 4096 instead of 16 lets the model ramble past the letter the parser expects.
6	`FRAME_FACTOR` rounding	1	Frame counts round down to a multiple of two, so a run may use fewer frames than requested.
7	prompt text / subtitles	4	Wording changes behaviour; the VideoMME preamble names subtitles even on the variant that passes none.
8	`temperature`	6	Any value above zero makes the run non-reproducible.
9	video reader backend	1	torchcodec, decord and torchvision land on marginally different frame indices.
10	model snapshot / `chat_template`	5	Different weights or template change the tokenisation of every vision block.